

MENTORA ACADEMY

EXCELLENCE IN EDUCATION

Edexcel International A Level Chemistry (WCH14)

UNIT 4 — TOPICAL WORKSHEET PACK: 14A.4

ANALYSING DATA FROM PH MEASUREMENTS

CANDIDATE NAME:	Usman
SUBJECT & CODE:	Chemistry Unit 4 — Rates, Equilibria & Organic (WCH14)
TOPIC AREA:	14: Acid-Base Equilibria
SUB-TOPIC:	14A.4: Analysing Data from pH Measurements
TOTAL QUESTIONS:	1 Questions (Structured + MCQs + Calculations)
TOTAL MARKS:	2 Marks
EXAMINER FAQs:	1 Empirical Examiner FAQs & Traps Included

HOW TO USE THIS TOPICAL PACK:

1. Complete all structured questions under timed exam conditions.
2. Pay close attention to command words (Explain, Deduce, Calculate, State).
3. Review the complete worked mark scheme in Section C.
4. Study the Empirical Examiner FAQs in Section D to avoid common candidate traps.

SECTION A: TOPICAL PRACTICE QUESTIONS — 14A.4

Q1. Calculate the pH of 0.1 mol dm⁻³ ethanoic acid ($K_a = 1.7 \times 10^{-5}$ mol dm⁻³).

[2 Marks]

WCH14/01/Oct20/Q19(c)

SECTION B: WORKED MARK SCHEME — 14A.4

Q	MARKING SCHEME & ACCEPTABLE ANSWERS	MARKS
Q1	Q1: Calculate the pH of 0.1 mol dm ⁻³ ethanoic acid ($K_a = 1.7 \times 10^{-5}$ mol dm ⁻³). $[H^+] = \sqrt{K_a \times [HA]} = \sqrt{1.7 \times 10^{-5} \times 0.1} = 1.30 \times 10^{-3}$. pH = 2.89	[2]

SECTION C: EMPIRICAL EXAMINER FAQs & COMMON TRAPS — 14A.4**FAQ #1: When can I use the approximation $[HA]_{eq} \approx [HA]_{initial}$?****Category: Examiner Trap & Technique****COMMON EXAMINER TRAP / MISTAKE:**

When can I use the approximation $[HA]_{eq} \approx [HA]_{initial}$?

MODEL ANSWER / EXAMINER EXPECTATION:

When dissociation is very small (usually when K_a is small and concentration is relatively high).